Now, there may be any number of salts which might be created and when you multiply that with the number of algorithms which can be used and then multiply it with the number of passwords that a user can input, the number is quite huge.

The biggest advantage of the salt is that you cannot create a file which contains a pre-computed set of hashes based on the user input. In case of Windows, it uses an algorithm which will embody the same hash when same passwords are entered. In such a condition, it is possible to produce a file which will contain the hashes which have been already calculated for every possible combination of characters in a password. Such a file is often called rainbow table. Linux defeats the possibility of 'rainbow table' with 'salt'.

Reset your forgotten password

Now let us come back to the point and have a look at the technique which allows you to reset your Linux user password. We will show you how to change the password of the 'root' user in this guide. Since Linux stores the password in a way that you can crack the password by using only the 'brute force' technique which is impractical due to its slow speed, we need to find a better way.

If you think carefully, it becomes obvious that the password can be changed if you can edit the password value field of a user in the /etc/shadow file. The problem is that you must be able to generate a proper password value with the right salt and hash. This can only be done when you already have a Linux distribution at hand! We will use the same technique to change the password of the Linux system:

1. Get your hands on a Live Linux distribution. It is recommended that you get Backtrack. The reason why we do not ask you to get an Ubuntu or Knoppix is that these may not allow you to have root access on the Live system, whereas undoubtedly Backtrack will, and by default. All further steps are written assuming that you boot into the live system as 'root'.
2. Boot the live media (USB or DVD) and wait until your desktop is ready. Remember that you need to select the graphical mode boot when booting into Backtrack. It is not mandatory but if you are using the command line mode, then you must follow the following steps carefully with corresponding actions.
3. It is time you open the password file of the installed Linux OS which is stored on the disk. You will later edit this file and store it back to reset the password. So first you'll have to locate the partition on which Linux root file system (/) is installed and then mount it.
   Mounting is typically done by using the mount command. So if your partition is /dev/sda2, then you should type this command on the shell:
   `mkdir /tmpmnt; mount /tmpmnt /dev/sda2`
   This will make the contents of the Linux partition available in the /tmpmnt directory.
   You should now open the /tmpmnt directory in a file browser such as Konqueror and go to the 'etc' directory and open the 'shadow' file. It will open in a text editor such as kwrite.
4. Now, you are ready to generate a new password. This is very simple. All you have to do is to open the terminal and type the command:
   `passwd`
   This will ask you to enter a password and ask again to confirm it. Enter it and make sure you remember what you enter. This will produce the password hash for the 'root' user on the live system. Since the system is live, nothing is saved on the disk and you need not worry!
5. Our next step is to reset the password of the installed Linux system. To do this, you've got to open the /etc/shadow file of the live system (do not open the same file you opened in step 3). In this file, search for the line beginning with the word 'root'. Once you have found it, copy the password value from this file and paste it at the corresponding place in the file which you opened in step 3; make sure that you have replaced the password value. If the line for the root user in the file on disk (which you opened in step 3) was
   `root:$1$fC214jQK$.fY88wY/w7J41CinaszNP1:14820:0:99999:7:::`
   and the line for the root user in the /etc/shadow file from the live system after changing the password (which you opened in this step) is
   `root:$1$g63hQ92U$h2ks8c.6p1qq.10d.wwQFr:14913:0:99999:7:::`
   Then, after this step, the line in the disk file should look like:
   `root:$1$g63hQ92U$h2ks8c.6p1qq.10d.wwQFr:14820:0:99999:7:::`
   You can use the same password value for other accounts too which you need to reset the password of. Remember that here we are resetting the password of the root user of the installed Linux system. If you wanted to change the password for some user other than root (on systems like Ubuntu, where root user is rarely used), then you need to replace the password value of that user.
6. Bingo! It's all done. Now you should save the file and reboot the system into the installed copy of Linux OS.
7. When the system asks for the password, you must enter the password which you had entered in step 4.

Now, you have unlocked your Linux system. Time to rejoice, isn't it? Yes, but once again, if it looks easy to you, it will look easy for an attacker too. Someone who can access your computer might try to use this technique to change or add a new user, change your password permanently or temporarily. It is indeed a great security risk and we are not even talking about networks. There is nothing much to worry, however. There are a few simple steps which you can take so that no one is able to use this technique to make things worse for you. Just follow the simple guidelines given below:

1. Disable booting from USB and DVDs from BIOS.
2. Lock down the computer using BIOS passwords. You might want to use an administrator password in BIOS so that no one can change the settings.
3. If you think the above methods cannot protect you because one can open the cabinet, remove the battery and reset the BIOS password, then the best defense would be to encrypt all the volumes (partitions) on Linux which contain sensitive data as well as the root partition itself (most Linux distributions support it, if you have a separate /boot partition).
4. If you are using GRUB to boot into Linux, it is a great idea to use a boot-loader password as well!

We hope that you will never worry about a forgotten Linux password now. Oh, will you?